Quantum Mirror™

Genesis OS Master Blueprint - Revised Current-System Edition

Build-ready specification for premium AI reflection, governed memory, Human Emotion Genome extraction, consent-gated profile enrichment, and production runtime architecture.

Purpose: Replace older Quantum Mirror references with the current workbook-governed Genesis OS stack.
Authority: Use with the workbook-aligned blueprint, score contract, and current workbook. Where any older quantum or scoring text conflicts, the workbook-governed stack wins.
Audience: Founders, product, backend, mobile, AI/ML, data, analytics, trust & safety, QA, and licensing partners.

Major revision note: this edition keeps Quantum Mirror detailed, but aligns it to the current Genesis OS system: workbook-governed score law, 52/23/13/12 score stack, Polarity v10 chemistry, contradiction clusters, Human Emotion Genome, evidence bands, consent scopes, and service boundaries.

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1. Executive Concept and Product Positioning

Quantum Mirror is the premium private intelligence layer of Genesis OS. It is not a generic chatbot bolted onto Relationship Scores. It is a governed clarity system that can explain scores, decode pattern loops, draft messages, interpret conflict, map triggers, and progressively deepen the user's emotional blueprint through structured extraction.

The premium value is context-aware guidance grounded in the user's assessment data, compatibility context, longitudinal behavior, reflection history, emotional state logs, contradiction clusters, companion-system telemetry, and evidence-banded forecast signals.

Primary design rule: the user experiences elegant reflection; the system underneath remains deterministic, versioned, consent-gated, cost-controlled, and auditable.

What the user feels	What the system actually does
Private, premium, personal	Checks entitlement, memory scope, rate limits, and context depth before session open.
Continuity	Reassembles bounded context from approved memory, structured signals, recent state, and consent-permitted partner context.
Precision	Uses evidence-band ceilings and canonical tags, not raw emotional dumping.
Safety	Routes abuse, self-harm, coercion, stalking, or

	manipulation signals into moderation flow with refusal/escalation logic.
Growth	Converts narrative interactions into versioned extracted signals, approved memories, and governed profile enrichment.

2. What Quantum Mirror Is and Is Not

It is	It is not
A context-aware reflection and guidance assistant powered by Genesis OS data.	Not therapy, diagnosis, legal advice, medical advice, or emergency care.
A premium layer for score interpretation, pattern translation, message help, post-date reflection, and guided self-awareness.	Not a truth oracle about a non-consenting partner.
A structured data collection surface that can ask targeted questions about history, trust ruptures, values, fears, grief, and growth.	Not a freeform memory bucket that stores everything forever.
A governed insight engine that converts narrative conversation into canonical tags, metrics, and evidence-banded suggestions.	Not a place where one dramatic session mutates slow identity or core scoring truth.
A Human Emotion Genome intake surface that captures signal families the assessment cannot fully reach.	Not an unmetered cost sink or unrestricted surveillance tool.

Hard non-negotiables

Deterministic first: visible scores and durable profile states must trace back to versioned formulas, thresholds, and canonical fields.

Consent-gated memory: no partner-facing insight may be produced from non-consented private data.

Evidence bands: every strong statement carries an internal evidence level and confidence basis.

Layer separation: slow identity, medium-horizon patterning, fast emotional state, and narrative memory must not collapse into one blob.

Auditable extraction: every structured signal saved from chat stores provenance, evidence snippets, extractor version, time scope, and save reason.

3. System Authority, Precedence, and Relation to Genesis OS

Quantum Mirror is a governed companion intelligence layer. It is not the scoring authority. It may read, explain, enrich, and forecast within policy, but it does not overwrite canonical workbook law.

Artifact	Role	If conflict exists
Current workbook / workbook-governed scoring stack	Source of truth for score law, tracked metrics, pair quantum logic, overlays, and polarity dimensions.	Workbook wins.

Unified master blueprint / workbook-aligned engineering blueprint	Implementation doctrine for data structures, storage, services, and build order.	Use latest corrected blueprint.
Quantum Mirror blueprint	Premium AI product, memory, extraction, moderation, and runtime specification.	Must align to workbook-governed stack; cannot override scoring law.
Companion Systems and Missing Layers docs	Runtime telemetry, state, event, and enrichment inputs.	Can enrich but not rewrite core score law.

Current-system alignment rules

Quantum Mirror may read the current 52 / 23 / 13 / 12 stack and explain its outputs, but may not redefine it.

Polarity v10 is the current chemistry layer. Quantum Mirror may reference chemistry outputs and ask follow-up questions that improve explanatory depth, but must not invent alternate polarity math.

Contradiction clusters are governed rules, not vibes. Quantum Mirror may surface reflection prompts and collect corroborating evidence, but cluster trigger law remains deterministic.

Human Emotion Genome signals may enrich fast state, medium pattern, and narrative memory according to evidence thresholds and consent; they do not directly hard-edit slow identity from a single session.

4. Premium Product Tiers, User Journeys, and Conversation Modes

Tier	User sees	System limits	Primary goal
Free Preview	Score explanation, a few guided prompts, one light mirror check, upsell surfaces.	Small context window; no deep memory; no durable extraction persistence; lite model path.	Create curiosity and trust.
Quantum Mirror Premium	Full personal sessions, message drafting, reflection analysis, trigger mapping, post-date and past-relationship decode.	Moderate monthly cap; personal memory scope; governed extraction; standard model path.	Recurring subscription revenue.
Quantum Mirror Couple	Dual-consent couple sessions, repair prompts, shared summaries, selected co-analysis.	Requires dual consent; stricter moderation; shared-memory rules; capped co-analysis depth.	Higher-value couple mode.
Enterprise / Licensed API	White-labeled partner access under contract.	Contractual rate limits; isolated tenant store; no direct user memory by default.	B2B expansion.

Primary conversation modes

Mode	Why user enters	Inputs read	Outputs returned

Score Deep Dive	User asks what a score, archetype, shadow, cluster, or match score means.	Forecast, compatibility, chemistry, archetype, shadow, active themes, evidence bands, trends.	Explanation, risk areas, next steps, links to reflections.
Past Relationship Decode	AI interviews user about a past relationship to collect narrative data and lessons.	History intake, trigger themes, betrayal patterns, ending style, grief residue.	Structured history extract, empathy summary, growth signals.
Message Studio	User wants help wording or interpreting a message.	Current state, match context, triggers, communication style, evidence band, consent scope.	Draft suggestions, tone options, caution notes.
After the Date / After the Conflict	User wants to interpret a recent interaction.	Event log, check-in state, perception-vs-reality markers, trigger history.	Grounded reflection, follow-up questions, state update.
Pattern Check	User asks why the same thing keeps happening.	Longitudinal trends, repeated tags, reflection clusters, history themes.	Pattern summary, evidence basis, intervention ideas.
Couple Reflection	Both parties consent to shared review.	Dual profiles, shared events, state deltas, consent scope, evidence band.	Joint summary, repair prompts, no unilateral exposure.

5. Current-System Score Context Quantum Mirror May Read and Explain

Quantum Mirror must understand the current Genesis OS score stack well enough to explain it, ask clarifying questions, and route explanatory logic without becoming the formula owner.

Layer	Current role	Quantum Mirror may do	Quantum Mirror may not do
52% Core pair quantum	Answer-pair interaction and deterministic compatibility backbone.	Explain outcomes; ask for narrative corroboration; produce evidence-banded summaries.	Change base weights or override pair rules from chat.
23% Behavioral / relational support	Archetype, shadow, communication, regulation, and related structured inputs.	Translate what these layers mean in human language.	Invent alternate shadow scoring or change archetype math from one session.
13% Chemistry layer	Polarity v10 chemistry: 8% polarity fit plus 5% pursuit/receptivity fit.	Explain chemistry tension, rhythm, and likely drift patterns.	Replace polarity law with freeform masculinity/femininity talk.
12% Overlay / adjustment lane	Forecast, contradiction effects, context, and approved system	Add explanatory context and route follow-up questions.	Override workbook-governed caps and penalties.

	adjustments.		

Public score family	What Quantum Mirror can say	Internal handling rule
Forecast / Readiness-facing 0-100 view	Explain signals, strengths, blind spots, and what would move the number over time.	Use evidence bands; avoid pretending one session permanently changes a score.
Compatibility 350-850	Explain why a pair lands in a tier, where attraction is strong, and where risk lives.	Use deterministic outputs and approved translation ranges.
Chemistry / Polarity subscore	Explain initiation, receptivity, tension, and drift implications.	Do not oversell chaos as destiny.

6. Data Capture Funnel: What to Ask, Store, Ignore, and Defer

Not every utterance becomes memory. Not every memory becomes a canonical field. Not every canonical field can influence scoring.

Layer	What it contains	Retention behavior	Allowed influence
A. Ephemeral session context	Recent messages, current intent, visible scores, last state logs, current partner context if allowed.	Short TTL; may expire after session or brief extension.	Answer generation only.
B. Session memory artifacts	Notable facts, preferences, narrative anchors, user-approved reminders.	Durable if within scope and product value.	Future continuity and user experience.
C. Structured profile signals	Canonical tags, metrics, history objects, counts, and decayed trends.	Saved only when schema and evidence rules are met.	Analytics, forecasting, medium-pattern enrichment, and approved downstream AI.
D. Moderation evidence	Abuse, coercion, self-harm, stalking, threats, panic, harassment evidence.	Separate safety retention policy.	Safety and trust operations only.

Data Quantum Mirror should explicitly avoid saving as durable product memory

Incidental gossip about unrelated third parties.

Sensitive trauma details that do not improve product value or safety handling.

Speculative diagnoses about self or others.

One-off spikes that have not repeated and do not meet extraction thresholds.

Anything outside consent scope or retention setting.

7. Human Emotion Genome: Missing Data Categories, Capture Timing, Labels, and Routing

Quantum Mirror is one of the main intake surfaces for the Human Emotion Genome. The assessment captures structured declarations; Quantum Mirror captures narrative-grade signal families that the assessment alone cannot fully reach.

Missing data family	What is missing	When to capture	How to capture	How to label / track	What Genesis OS should infer
Trigger activation specifics	What exact cues create body alarm and speed of escalation.	Post-date, post-conflict, pattern-check, or guided trigger mapping.	Targeted prompts plus recent-event interpretation.	trigger_tags[], trigger_intensity_0_1, trigger_latency_class, evidence_band.	Fast-state sensitivity, medium-pattern risk, follow-up intervention prompts.
Repair and re-entry behavior	How user returns after rupture, delay tolerance, apology acceptance, deactivation length.	After conflict, couple reflection, past relationship intake.	Behavioral recounting plus structured follow-up.	repair_style_tag, reengage_delay_bucket, rupture_repair_ratio.	Forecast stability, couple guidance, contradiction checks.
Boundary expression style	How user says no, asks for space, requests reassurance, or exits.	Message Studio, growth check-in, conflict review.	Drafting assistance and reflection prompts.	boundary_expression_tags, boundary_clarity_score, boundary_follow through_score.	Communication coaching and forecast quality.
Narrative pain themes	Recurring themes across relationships not fully visible in answer bank.	Past relationship decode, pattern check.	Long-form story intake with extractor validation.	history_theme_tags[], theme_counts, unresolved_theme_score.	Medium-pattern enrichment, reflection clusters, grief routing.
Perception-vs-reality gap	Difference between objective event and emotional interpretation.	After the date/conflict, event-linked sessions.	Ask for timeline, exact events, and internal meaning separately.	pvrd_gap_score, objective_event_count, interpretation_intensity.	State calibration, overread/underread detection, coaching tone.
Reassurance and communication preferences	Wording and pacing that helps or hurts regulation.	Message Studio and live reflection sessions.	Preference prompts; feedback on draft suggestions.	preferred_coaching_tone, communication_comfort_tags, reassurance_preference_tags.	Personalization and message generation quality.
Grief residue and closure style	How old relationships still impact state and	Past relationship intake, anniversaries,	History entry creation plus check-ins.	grief_residue_score, ending_style_tag,	Medium-pattern enrichment and intervention

	relational choice.	post-trigger revisits.		closure_status.	timing.
Growth evidence	What the user now handles better than before.	Periodic growth check-in.	Explicit compare-then-vs-now prompts.	growth_signal_tags, behavior_shift_counts, confidence_band.	Progress surfaces, coaching, retention.

Genome signal families to store

Family	Examples	Typical storage object
Tone and stance	hedged reflective, guarded, overexplaining, dry detached, self-blaming	qm_message.tone_profile_primary, qm_extracted_signal
Behavior pattern	pursuer, distancer, soft entry, preemptive withdrawal, appeaser	behavior_pattern_tag, pattern counters
Narrative themes	abandonment fear, emotional invisibility, overfunctioning, trust fracture	history_theme_tags, relationship_history_entry
Need and boundary	reassurance need, dignity-preserve boundary, pace boundary, privacy need	signal_family VALUE / BOUNDARY / RELATIONAL_NEED
State and activation	guarded hopeful, numb, activated, ashamed, drained	fast-state objects with decay windows
Risk and safety	coercion concern, stalking, panic, dependency risk	moderation event family, separate retention

8. Canonical Taxonomy for Words, Tone, Phrasing, Themes, Behaviors, and State

Words should not be stored as fuzzy vibes. Quantum Mirror maps user language into controlled labels so analytics, AI, and scoring speak the same internal language.

Source layer	Stored field	Example raw phrasing	Canonical label example	Extraction rule	Used for
Word choice	lexical_markers[]	I keep rereading every text	LEX_REREADS_MESSAGES	literal or near-literal phrase match	supporting evidence only
Tone framing	tone_profile.primary	Depends on the day honestly...	TONE_HEDGED_REFLECTIVE	hedge plus reflective wording pattern	response style and extraction confidence
Self-positioning	self_positioning_tag	I usually make myself smaller	SELF_MINIMIZING	first-person self-reduction cue	readiness coaching

Conflict phrasing	conflict_posture_tag	I joke first, then maybe say it	CONFLICT_SOFT_ENTRY	humor-as-entry plus delayed directness	communication style
Attachment cue	attachment_trigger_tag	late replies make my chest drop	TRIGGER_DELAYED_RESPONSE	body alarm plus delay reference	trigger mapping
Narrative theme	history_theme_tags[]	I keep staying too long	THEME_OVERSTAYING	repetition plus regret framing	past relationship analysis
Boundary signal	boundary_expression_tag	I do not want to beg to be chosen	BOUNDARY_DIGNITY_PRESERVE	self-protection statement	growth signal
Behavior inference	behavior_pattern_tag	I disappear before they can switch up	PATTERN_PREEMPTIVE_WITHDRAWAL	leave-first protection logic	forecasting
Repair cue	repair_style_tag	I come back when I cool off	REPAIR_DELAYED_REENGAGE	distance then reconnect	couple guidance
Risk cue	safety_flag	He kept showing up after I said stop	SAFETY_STALKING_CONCERN	clear coercive persistence	moderation escalation

Controlled enums

ToneProfile.primary: TONE_DIRECT, TONE_HEDGED_REFLECTIVE, TONE_WARM_OPEN, TONE_DRY_DETACHED, TONE_GUARDED, TONE_OVEREXPLAINING, TONE_CHAOTIC_ACTIVATED, TONE_FLAT_EXHAUSTED, TONE_SELF_BLAMING, TONE_ACCUSATORY.

PhrasingStyle.tags: PHRASE_JOKING_DEFLECTION, PHRASE_QUALIFIER_HEAVY, PHRASE_MIND_READING, PHRASE_ABSOLUTE_LANGUAGE, PHRASE_BODY_SIGNAL_REFERENCE, PHRASE_PAST_PATTERN_REFERENCE, PHRASE_GROWTH_OWNERSHIP, PHRASE_PEACEKEEPING, PHRASE_CONTROLLED_DISTANCE, PHRASE_REASSURANCE_SEEKING.

NarrativeTheme.tags: THEME_ABANDONMENT_FEAR, THEME_EMOTIONAL_INVISIBILITY, THEME_INCONSISTENCY_ADDICTION, THEME_OVERFUNCTIONING, THEME_SELF_ERASURE, THEME_LOYALTY_TO_WOUND, THEME_FIXING_OTHERS, THEME_SAFETY_CRAVING, THEME_CONTROL_AVOIDANCE, THEME_TRUST_FRACTURE, THEME_GRIEF_RESIDUE.

BehaviorPattern.tags: PATTERN_PURSUER, PATTERN_DISTANCER, PATTERN_PASSIVE_DRIFTER, PATTERN_SOFT_ENTRY, PATTERN_HYPER_VIGILANT_DECODER, PATTERN_PREEMPTIVE_WITHDRAWAL, PATTERN_OVEREXPLAINER, PATTERN_APPEASER, PATTERN_SCOREKEEPER, PATTERN_REPAIR_ORIENTED.

9. Core Database Schema and Field-Level Definitions

Quantum Mirror requires its own bounded schema family while reusing Genesis OS identity, scoring, state, event, and packet objects. Keep logic in database truth, not hidden in prompts.

9.1 quantum_mirror_subscription

Field	Type	Req.	Description	Notes / Rules
subscription_id	uuid	yes	Primary subscription	One active row per

			record	user per product tier.
user_id	uuid	yes	Links to user profile	FK users.user_id.
tier_code	enum	yes	FREE_PREVIEW / QM_PREMIUM / QM_COUPLE	Do not use free-text tier names.
billing_provider	enum	yes	APPLE / GOOGLE / STRIPE / PARTNER	Needed for entitlement sync.
billing_status	enum	yes	ACTIVE / PAST_DUE / CANCELED / TRIAL / PAUSED	Source of truth for entitlement.
period_start_at	timestamp	yes	Billing cycle start	UTC.
period_end_at	timestamp	yes	Billing cycle end	UTC.
session_quota	int	yes	Allowed premium sessions in period	May be null for unlimited plan if cost model allows.
message_quota	int	yes	Allowed messages in period	Soft or hard cap.
context_depth_tier	enum	yes	SHALLOW / STANDARD / DEEP	Controls memory and model path.

9.2 quantum_mirror_session

Field	Type	Req.	Description	Notes / Rules
qm_session_id	uuid	yes	Primary session id	Created when user enters Quantum Mirror.
user_id	uuid	yes	Session owner	FK users.user_id.
conversation_mode	enum	yes	SCORE_DEEP_DIVE / PAST_RELATIONSHIP / MESSAGE_STUDIO / AFTER_CONFLICT / PATTERN_CHECK / COUPLE_REFLECTION	Drives prompt path and extraction recipe.
partner_context_user_id	uuid	no	Optional partner or match context	Only if consent scope permits.
entry_surface	enum	yes	PAYWALL / PROFILE / MATCH / SANCTUARY / SCORE_RESULT /	Analytics and optimization.

			PUSH	
session_status	enum	yes	OPEN / CLOSED / ESCALATED / INTERRUPTED / RATE_LIMITED	Lifecycle control.
memory_scope	enum	yes	EPHEMERAL_ONLY / PERSONAL_MEMORY / PERSONAL_PLUS_COUPLE	Must obey consent.
model_path	enum	yes	LITE / STANDARD / DEEP_REASONING	Cost and performance tracking.
evidence_band_ceiling	enum	yes	LOW / MEDIUM / HIGH	Prevents overclaiming in low-data contexts.
started_at	timestamp	yes	Session start	UTC.
ended_at	timestamp	no	Session end	UTC.

9.3 quantum_mirror_message

Field	Type	Req.	Description	Notes / Rules
qm_message_id	uuid	yes	Primary message id	
qm_session_id	uuid	yes	Parent session	
speaker_role	enum	yes	USER / ASSISTANT / SYSTEM / MODERATOR	
message_index	int	yes	Order in session	Monotonic per session.
message_text	text	yes	Normalized message body	Retain exact user text subject to deletion policy.
redaction_state	enum	yes	RAW / REDACTED / MASKED	For safety and privacy.
detected_intent_primary	enum	no	Top user intent	Stored after classification.
tone_profile_primary	enum	no	Primary tone tag	Stored after tone analysis.
safety_risk_level	enum	no	NONE / LOW / MODERATE / HIGH /	Moderation pipeline.

			CRITICAL	
created_at	timestamp	yes	Timestamp	UTC.

9.4 qm_extracted_signal

Field	Type	Req.	Description	Notes / Rules
signal_id	uuid	yes	Primary extracted signal row	
user_id	uuid	yes	Owner of signal	
source_qm_session_id	uuid	yes	Origin session	
source_qm_message_id	uuid	yes	Origin message	May also point to system interview response.
signal_family	enum	yes	TRIGGER / HISTORY_THEME / COMMUNICATION_STYLE / VALUE / BOUNDARY / RELATIONAL_NEED / GROWTH_SIGNAL / SAFETY_SIGNAL / PREFERENCE / STATE	Canonical family.
signal_code	varchar	yes	Controlled enum such as THEME_SELF_ERASURE	Must exist in registry.
signal_value_type	enum	yes	TAG / SCORE / BOOLEAN / TEXT_SHORT / COUNT / DATE_APPROX	
signal_value	json	yes	Typed payload	Example {tag:'THEME_SELF_ERASURE'} or {score:0.74}.
time_scope	enum	yes	CURRENT / PAST_PATTERN / HISTORICAL_RELATIONSHIP / STABLE_TRAIT / UNKNOWN	Prevents tense confusion.
evidence_band	enum	yes	LOW / MEDIUM /	Internal certainty.

			HIGH	
extractor_version	varchar	yes	Version of extractor recipe / prompt / model	
save_status	enum	yes	PROPOSED / APPROVED_AUTO / APPROVED_REVIEWED / REJECTED / EXPIRED	Governance.
provenance_snippet	text	yes	Short support span	Cap length and scrub sensitive data if needed.

9.5 relationship_history_entry

Field	Type	Req.	Description	Notes / Rules
history_entry_id	uuid	yes	Primary past relationship record	
user_id	uuid	yes	Owner	
relationship_label	varchar	no	User-facing label such as 'college ex'	Avoid legal names when unnecessary.
relationship_type	enum	yes	DATING / COMMITTED / MARRIAGE / SITUATIONSHIP / TALKING / OTHER	
approx_start_date	date	no	Approximate start	Allow month/year granularity.
approx_end_date	date	no	Approximate end	Allow null if ongoing.
most_salient_themes	json	yes	Array of canonical theme tags	Derived from intake and later sessions.
conflict_posture_tag	enum	no	Primary conflict style in that relationship	
ending_style_tag	enum	no	How relationship ended	MUTUAL / GHOSTED / LEFT / WAS_LEFT / CYCLICAL / UNKNOWN.
grief_residue_score	decimal	no	0.00-1.00 unresolved impact score	Do not show raw decimal unless

				formatted.
growth_lessons_summary	text	no	Short reflective summary	Can be AI-written and user-editable.

9.6 qm_memory_item

Field	Type	Req.	Description	Notes / Rules
memory_item_id	uuid	yes	Durable memory row	
user_id	uuid	yes	Owner	
memory_type	enum	yes	PREFERENCE / HISTORY_ANCHOR / GOAL / BOUNDARY / COMM_STYLE / IMPORTANT_CONTEXT	
memory_text	text	yes	Human-readable memory statement	User can review/delete.
source_signal_ids	json	yes	Supporting signal ids	No orphan memories.
visibility_scope	enum	yes	PRIVATE_USER / PRIVATE_SYSTEM / COUPLE_SHARED	Must obey consent.
retention_class	enum	yes	STANDARD / SHORT / REGULATED / SAFETY	Drives deletion behavior.
review_state	enum	yes	AUTO / USER_CONFIRMED / USER_REJECTED / REVIEWED	

9.7 qm_helpfulness_feedback

Field	Type	Req.	Description	Notes / Rules
feedback_id	uuid	yes	Primary feedback row	
qm_session_id	uuid	yes	Session being rated	
qm_message_id	uuid	no	Specific answer rated	Optional.
user_id	uuid	yes	Rater	
helpfulness_score	int	yes	1-5	

clarity_score	int	no	1-5	
felt_seen_score	int	no	1-5	Optional premium signal.
too_generic_flag	bool	yes	Whether answer felt generic	
too_confident_flag	bool	yes	Whether answer overclaimed	Important calibration signal.
free_text_feedback	text	no	Optional comment	

10. Session Memory, Extraction, Profile Mutation, and Contradiction Handling

This is the heart of the system. Quantum Mirror becomes valuable when it converts reflective conversation into governed long-term intelligence without contaminating slow identity with every passing mood.

Layer	Examples	Who can update it	Update rule
Slow identity	Archetype, shadow distribution, baseline readiness inputs, core stable preferences	Assessment engine, governed jobs, explicit user edits where allowed	Never changed directly by one Quantum Mirror session.
Medium pattern layer	Trigger intensities, repeated themes, communication tendencies, grief residue trend, contradiction corroboration	Batch updater plus high-evidence extraction jobs	Needs repeated evidence or corroboration.
Fast state layer	Current emotional state, post-conflict activation, post-date hopefulness, current stress	Immediate session and state-check services	May update per event/session and should decay.
Narrative memory	Past relationship entries, durable goals, boundaries, user-approved memories	Quantum Mirror extraction plus user confirmation	Editable and deletable by user.

Extraction recipe

Step 1: classify session intent and identify extraction recipe family.

Step 2: run lightweight parsers for tone, lexical markers, temporal references, and safety cues.

Step 3: run structured extractor prompt against bounded context and canonical registry.

Step 4: produce candidate signal objects with evidence snippets, time scope, and confidence score.

Step 5: apply deterministic validator: allowed enum, consent scope, evidence threshold, contradiction check, duplicate suppression, and profile mutation policy.

Step 6: save as PROPOSED or APPROVED according to policy; high-impact identity changes always route through slower governed jobs.

Step 7: optionally ask the user to confirm durable memories in plain language.

Step 8: log extraction metrics for review, false-positive analysis, and future tuning.

Contradiction and reflection handling

Cluster type	Quantum Mirror role	What gets saved	What does not happen
Locked active clusters such as Vulnerability Loop	Ask corroborating questions, provide reflection prompts, explain tension between signals.	cluster observation event, evidence snippets, user response, maybe medium-pattern support signal.	Does not rewrite deterministic cluster trigger law.
Proposed or emerging clusters	Collect repeat evidence and mark as proposed patterns.	proposed signal rows with LOW/MEDIUM evidence.	Does not auto-promote speculative cluster truth to profile.
Pair contradictions	Explain mismatched rhythm or meaning when consent allows pair context.	pair reflection note, not raw partner exposure.	Does not reveal unilateral private data.

11. AI Orchestration, Evidence Bands, Refusal Logic, and Output Schemas

Quantum Mirror should be built as a pipeline, not a single monolithic prompt.

Component	Role
Intent classifier	Maps each user turn to approved intents and sub-intents.
Tone / stance detector	Identifies tone, activation, self-positioning, and conversational posture.
Safety detector	Finds self-harm, abuse, coercion, stalking, panic, crisis, or high-risk dependency language.
Structured extractor	Turns narrative content into canonical signal candidates.
Guidance generator	Produces the user-visible answer using evidence-banded context.
Helpfulness calibrator	Scores whether output felt useful and adjusts future prompt selection.
Cost router	Chooses lite vs standard vs deep path based on entitlement, complexity, and urgency.

Band	Internal meaning	Allowed language	Example
LOW	Sparse data, one-off	Use soft framing: might, it	It sounds like delayed

	mention, or contradictory evidence	sounds like, one possibility.	replies might be a trigger for you, but I would want more history before treating that as settled.
MEDIUM	Repeated cues or multi-source support but not stable enough for hard claims	Use moderate framing: seems to, often, there is a pattern suggesting.	There is a pattern suggesting that mixed signals often push you into overanalysis.
HIGH	Strong multi-source support over time or validated by user confirmation	Use clear but bounded framing: consistently, your data shows, you tend to.	Across your assessment and recent reflections, you consistently protect yourself by getting quieter before conflict fully opens.

Refusal and redirection classes

Clinical diagnosis request: refuse diagnosis, offer reflective framing, encourage professional help where appropriate.

Emergency or self-harm: exit normal mode and provide crisis-oriented guidance and escalation per policy.

Requests to surveil or manipulate another person: refuse and redirect toward self-awareness and respectful communication.

Legal, medical, or financial claims outside scope: refuse specialized advice and state limits.

Non-consented partner exposure: explain that the system cannot expose private partner data without consent.

Harassment or abuse planning: refuse and escalate if required by safety policy.

12. API Contracts, Service Boundaries, Packet Model, and Event Model

Quantum Mirror needs clean service boundaries so developers do not bury logic inside controllers or prompts.

Service	Responsibility
Entitlement Service	Verifies subscription status, quotas, and context depth tier.
Context Assembly Service	Composes Quantum Mirror packet from Genesis OS stores and consent scopes.
Conversation Service	Owns session and message lifecycle.
AI Orchestration Service	Classifier, generator, extractor, and post-process pipeline.
Signal Validator Service	Checks canonical enums, thresholds, contradictions, and save policy.
Memory Service	Stores, deletes, versions, and invalidates memory items.

Moderation Service	Detects risk, queues reviews, and triggers escalation.
Analytics Service	Logs usage, cost, helpfulness, and safety metrics.

Core request / response contracts

POST /qm/session/start -> request { user_id, conversation_mode, entry_surface, partner_context_user_id | null } response { qm_session_id, session_status, entitlement_tier, remaining_message_quota, context_depth_tier, evidence_band_ceiling }

POST /qm/message/send -> request { qm_session_id, user_message, client_timestamp } response { assistant_message_id, assistant_text, follow_up_question | null, suggested_actions[], evidence_band_used, safety_outcome, memory_save_prompts[] }

POST /qm/memory/confirm -> request { memory_prompt_id, decision: CONFIRM | REJECT } response { status, memory_item_id | null }

GET /qm/profile-summary -> response { user_id, readiness_or_forecast_score, archetype_primary, shadow_primary, top_trigger_tags[], active_theme_tags[], recent_state }

Packet model blocks

Packet block	Contains	Source	Used by
Identity summary	user id, tier, consent scope, entitlement snapshot	User and subscription services	Orchestrator, gatekeeper
Blueprint summary	forecast/readiness-facing view, compatibility context if relevant, archetype, shadow, top flags	Score services	Score explanation, follow-up questioning
State block	recent state, decays, last check-ins, active trigger load	State service	Response planning
Memory block	approved personal memories, history anchors, growth notes	Memory service	Continuity and extraction
Safety block	active risk flags, moderation holds, escalation notes	Moderation service	Safety shaping and refusal
Partner context block	consented pair summary only	Pair context service	Couple or match-aware flows

Event model

Field	Type	Description
event_id	uuid	Global unique id.
event_type	enum	QM_SESSION_STARTED / QM_MESSAGE_SENT / QM_SIGNAL_PROPOSED /

		QM_MEMORY_CONFIRMED / QM_SAFETY_ESCALATED / QM_RESPONSE_RATED / etc.
user_id	uuid	Actor or owner.
subject_user_id	uuid	Optional second-party subject.
qm_session_id	uuid	Optional session link.
payload	json	Typed event body.
version	varchar	Event schema version.
created_at	timestamp	UTC.

13. Consent Architecture, Deletion, Retention, Moderation, and Training Controls

Scope code	Meaning	Allows save of	Does not allow
CONSENT_QM_EPHEMERAL	Use conversation only for current answer.	Session context.	Durable memory or profile extraction.
CONSENT_QM_PERSONAL_MEMORY	Use and save personal memories and low-risk extracted signals.	Personal memory, stable preferences, history themes.	Couple/shared memory.
CONSENT_QM_PROFILE_ENRICHMENT	Allow approved extracted signals to enrich Genesis OS personal profile.	Trigger tags, pattern tags, history entries.	Direct mutation of slow identity without rules.
CONSENT_QM_COUPLE_SHARED	Allow selected shared summaries for couple mode when both users agree.	Shared session summaries, agreed tasks.	Unilateral private disclosures to partner.
CONSENT_QM_TRAINING_AGGREGATE	Allow anonymized aggregate use for tuning and analytics.	De-identified metrics.	Raw identifiable chat exports for model training without separate policy.

Deletion and retention principles

User must be able to view and delete durable Quantum Mirror memories and past relationship entries subject to legal and safety exceptions.

Deleting a memory item should remove it from future context assembly immediately and queue downstream cache invalidation.

Derived structured signals tied only to deleted content should be soft-deactivated or deleted according to retention class and policy.

Moderation evidence may retain longer than product memory where required for abuse handling or dispute resolution.

Training use must rely on de-identified aggregates unless separate policy and consent exists.

Training and governance guardrails

Do not train future models on raw identifiable user chat by default.

Use aggregate metrics, label confusion matrices, helpfulness deltas, and extractor false-positive analysis first.

Store provenance for every extracted signal used in quality review.

Keep a versioned registry of extractor recipes, validators, enums, and evidence-band thresholds.

14. Cost Controls, Quotas, Analytics, QA, and Launch Path

Control	Implementation rule	Why it matters
Model path routing	Use LITE, STANDARD, or DEEP_REASONING by entitlement, complexity, urgency, and safety need.	Protects margin and latency.
Context depth tier	SHALLOW, STANDARD, DEEP determines memory amount and packet size.	Prevents unnecessary token burn.
Message/session quotas	Cap monthly usage per tier and log remaining quota at gate decisions.	Avoids premium becoming a compute fire.
Evidence-band ceiling	Limit claim strength in low-data contexts.	Protects trust and reduces hallucinated certainty.
Extraction gating	Separate answer generation from save approval.	Prevents profile contamination.
Helpfulness feedback loop	Collect too_generic and too_confident feedback.	Improves product quality with real guardrails.

Narrow-v1 launch path

Launch with SCORE_DEEP_DIVE, PAST_RELATIONSHIP, MESSAGE_STUDIO, AFTER_CONFLICT, and PATTERN_CHECK before couple-shared mode.

Keep extraction to a smaller approved registry at first: trigger tags, history themes, communication preferences, growth signals, boundary tags, and selected state objects.

Require confirmation for durable memory creation during early rollout.

Do not let Quantum Mirror directly alter archetype or shadow assignments in v1.

Turn on couple-shared mode only after dual-consent flows, moderation, and unilateral-disclosure protections are verified.

15. Build Order and Engineering Implementation Checklist

Phase	Build first	Why first	Done when
Phase 1	Entitlement service, session/message schema,	Foundation and auditability.	A user can open a gated session, send a message,

	event model, context assembly skeleton.		and log events safely.
Phase 2	Intent classifier, tone detector, safety detector, evidence-band ceilings, answer generation pipeline.	Safe basic product loop.	User gets bounded answers without durable memory mutation.
Phase 3	Structured extractor, validator, memory prompts, qm_memory_item CRUD, deletion flows.	Governed continuity.	Durable memory can be proposed, confirmed, deleted, and invalidated.
Phase 4	History intake, trigger mapping, growth check-ins, relationship_history_entry, genome signal capture.	Missing-layer data collection.	Narrative-grade signals flow into approved objects.
Phase 5	Packet enrichment from current Genesis OS stack, contradiction support, approved profile enrichment jobs.	Current-system alignment.	Quantum Mirror reads and explains current stack cleanly.
Phase 6	Helpfulness calibration, analytics dashboards, quota tuning, couple mode preparation.	Operational maturity.	Product can be scaled with margin and safety visibility.

Developer checklist

Implement all schema objects before prompt tuning.

Version every prompt, extractor, validator, and packet shape.

Separate answer-generation path from save path.

Never mutate slow identity directly from a single Quantum Mirror session.

Keep partner context behind explicit consent gates.

Provide user-visible memory management controls.

Log token counts, cost estimate, evidence band, and safety outcome for every answer.

Route abuse and coercion signals into moderation storage, not product memory.

16. Appendices

Appendix A - Example end-to-end flow

1. User opens After the Conflict mode from Sanctuary.

2. Entitlement and consent are checked; session opens with STANDARD depth and MEDIUM evidence ceiling.

3. Context assembly pulls recent state, last two state logs, approved trigger tags, partner context if allowed, and current score summary.

4. User describes the event. Tone detector marks TONE_GUARDED plus PHRASE_BODY_SIGNAL_REFERENCE.

5. Extractor proposes TRIGGER_DELAYED_RESPONSE, PATTERN_OVEREXPLAINER, and pvrd_gap_score 0.62 with LOW/MEDIUM evidence.

6. Validator approves state update, leaves pattern tag as PROPOSED, and prompts user to confirm a memory only if useful.

7. Assistant returns grounded reflection with moderate language, a follow-up question, and no overclaiming.

8. Event logs record model path, prompt version, token counts, evidence band, extraction outcomes, and feedback hooks.

Appendix B - Quantum Mirror implementation notes tied to the current system

Quantum Mirror reads the current score stack as explanation context, not formula authorship.

Human Emotion Genome signals should enter Genesis OS through approved packet routes and update lanes.

Companion systems such as grief, drift, shadow activation, and state telemetry may enrich context packets, but their weights remain governed outside Quantum Mirror.

Polarity v10 remains the chemistry law; use it for explanation and follow-up prompts, not for freeform redefinition.

Appendix C - Minimum quality gates before release

Gate	Requirement
Safety	Crisis, coercion, stalking, abuse, self-harm, and manipulation refusal paths tested.
Memory	User can confirm, reject, view, and delete memories; cache invalidation works.
Evidence	Output language obeys band ceilings and confidence rules.
Consent	No couple-shared or partner-facing content leaks from unilateral private data.
Cost	Quota exhaustion, path routing, and observability verified.
Audit	All sessions, outputs, extractions, and moderation actions are versioned and logged.

Appendix D - What Quantum Mirror sends to Genesis OS later

Quantum Mirror does not stream every message into Genesis OS core. It sends approved structured artifacts: confirmed memory items, approved extracted signals, relationship history entries, state deltas, and de-identified analytics events.

Recommended outbound packet blocks: signal_batch, state_delta, memory_delta, history_entry_delta, moderation_event_reference, analytics_summary, and provenance_manifest.

Every outbound artifact must include user_id, source_qm_session_id, source_qm_message_ids, extractor_version, validator_version, evidence_band, consent_scope, created_at, and retention_class.

Genesis OS should treat Quantum Mirror outbound data as enrichment input and route it through update-lane policy: fast-state writes may apply immediately, medium-pattern writes require thresholds, slow identity requires governed jobs only.

Outbound object	When sent	Destination	What it tells Genesis OS
signal_batch	End of turn or end of session after validation	Signal intake service	What canonical tags, scores, or counts were newly observed.
state_delta	Immediately after state-changing session or check-in	State service	How current activation, hopefulness, or guardedness changed and when to decay it.
memory_delta	After user confirms memory save	Memory and profile context service	What durable reminder should shape future context assembly.
history_entry_delta	After past-relationship intake or major edit	Relationship history store	What prior relationship themes, ending style, and grief residue now exist.
moderation_event_reference	Immediately on flagged content	Moderation service	What safety workflow was triggered and which conversation artifacts matter.
analytics_summary	Session close	Analytics warehouse	How often modes are used, what costs were incurred, and what felt helpful or overconfident.

Appendix E - Implementation anti-patterns to avoid

Do not let prompt text become the only place a rule exists. If the rule matters, encode it in schema, validators, or service logic.

Do not save raw emotional monologues as durable analytics labels. Save canonical labels with provenance, not diary dumps masquerading as structure.

Do not merge slow identity, medium pattern, and fast state into one profile blob. That is how the machine starts hallucinating personality evolution from a bad Tuesday.

Do not expose private partner inferences in couple mode without dual consent and explicit visibility rules.

Do not allow a single Quantum Mirror session to alter archetype or shadow assignments, final compatibility law, or workbook-owned weights.

Appendix F - Build tasks by discipline

Discipline	Must build	Input artifacts	Acceptance check
Backend	Session/message services, validator, memory CRUD, event log, moderation hooks	This blueprint, service architecture spec, data dictionary	Every response and save path is versioned and auditable.
AI/ML	Classifier, extractor, generator routing, evidence-band limiter, helpfulness loop	AI governance spec, feature engineering spec, current enums	Output language obeys band ceilings and save candidates match registries.
Mobile / Frontend	Session UI, premium gates, memory ledger, mode entry points, save confirmations	Product sections of this blueprint	User can clearly see what is saved, what is private, and what is shared.
Data / Analytics	Warehouse events, cost dashboards, false-positive review, tuning metrics	Event model, analytics summary, training controls	Team can monitor cost, safety, and quality without reading raw chat at scale.
Trust & Safety	Escalation queue, policy outcomes, retention exceptions	Moderation sections, consent scopes	High-risk sessions are contained and logged.